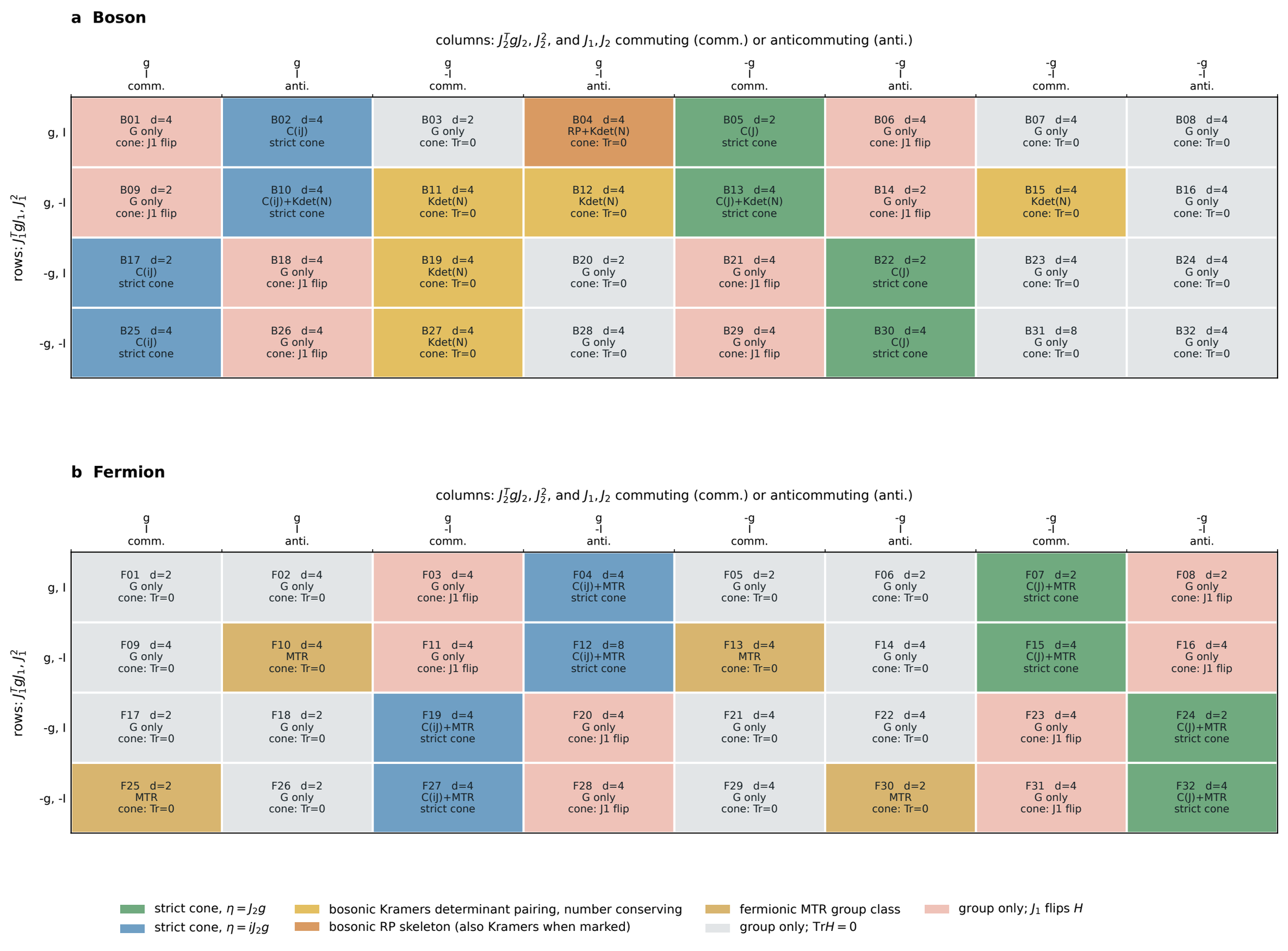
a Boson
columns: $J_2^T g J_2$, J_2^2, and J_1, J_2 commuting (comm.) or anticommuting (anti.)
rows: $J_1^T g J_1$, J_1^2
g I comm.
g I anti.
g -I comm.
g -I anti.
-g I comm.
-g I anti.
-g -I comm.
-g -I anti.
g, I
g, -I
-g, I
-g, -I
B01 d=4 G only cone: J1 flip
B02 d=4 C(iJ) strict cone
B03 d=2 G only cone: Tr=0
B04 d=4 RP+Kdet(N) cone: Tr=0
B05 d=2 C(J) strict cone
B06 d=4 G only cone: J1 flip
B07 d=4 G only cone: Tr=0
B08 d=4 G only cone: Tr=0
B09 d=2 G only cone: J1 flip
B10 d=4 C(iJ)+Kdet(N) strict cone
B11 d=4 Kdet(N) cone: Tr=0
B12 d=4 Kdet(N) cone: Tr=0
B13 d=4 C(J)+Kdet(N) strict cone
B14 d=2 G only cone: J1 flip
B15 d=4 Kdet(N) cone: Tr=0
B16 d=4 G only cone: Tr=0
B17 d=2 C(iJ) strict cone
B18 d=4 G only cone: J1 flip
B19 d=4 Kdet(N) cone: Tr=0
B20 d=2 G only cone: Tr=0
B21 d=4 G only cone: J1 flip
B22 d=2 C(J) strict cone
B23 d=4 G only cone: Tr=0
B24 d=4 G only cone: Tr=0
B25 d=4 C(iJ) strict cone
B26 d=4 G only cone: J1 flip
B27 d=4 Kdet(N) cone: Tr=0
B28 d=4 G only cone: Tr=0
B29 d=4 G only cone: J1 flip
B30 d=4 C(J) strict cone
B31 d=8 G only cone: Tr=0
B32 d=4 G only cone: Tr=0
b Fermion
columns: $J_2^T g J_2$, J_2^2, and J_1, J_2 commuting (comm.) or anticommuting (anti.)
rows: $J_1^T g J_1$, J_1^2
g I comm.
g I anti.
g -I comm.
g -I anti.
-g I comm.
-g I anti.
-g -I comm.
-g -I anti.
g, I
g, -I
-g, I
-g, -I
F01 d=2 G only cone: Tr=0
F02 d=4 G only cone: Tr=0
F03 d=4 G only cone: J1 flip
F04 d=4 C(iJ)+MTR strict cone
F05 d=2 G only cone: Tr=0
F06 d=2 G only cone: Tr=0
F07 d=2 C(J)+MTR strict cone
F08 d=2 G only cone: J1 flip
F09 d=4 G only cone: Tr=0
F10 d=4 MTR cone: Tr=0
F11 d=4 G only cone: J1 flip
F12 d=8 C(iJ)+MTR strict cone
F13 d=4 MTR cone: Tr=0
F14 d=4 G only cone: Tr=0
F15 d=4 C(J)+MTR strict cone
F16 d=4 G only cone: J1 flip
F17 d=2 G only cone: Tr=0
F18 d=2 G only cone: Tr=0
F19 d=4 C(iJ)+MTR strict cone
F20 d=4 G only cone: J1 flip
F21 d=4 G only cone: Tr=0
F22 d=4 G only cone: Tr=0
F23 d=4 G only cone: J1 flip
F24 d=2 C(J)+MTR strict cone
F25 d=2 MTR cone: Tr=0
F26 d=2 G only cone: Tr=0
F27 d=4 C(iJ)+MTR strict cone
F28 d=4 G only cone: J1 flip
F29 d=4 G only cone: Tr=0
F30 d=2 MTR cone: Tr=0
F31 d=4 G only cone: J1 flip
F32 d=4 C(J)+MTR strict cone
strict cone, $\eta = J_2 g$
strict cone, $\eta = iJ_2 g$
bosonic Kramers determinant pairing, number conserving
bosonic RP skeleton (also Kramers when marked)
fermionic MTR group class
group only; Tr$H = 0$
group only; J_1 flips H